Revisiting SIF abstraction rules with SPARQL for querying BioPAX

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Abstract

BioPAX (Biological PAthway eXchange) is a standard Semantic Web format for the representation of biological pathways. Its expressivity finely describes biological information, but the counterpart is a complexity hindering downstream analyses and reasoning tasks. Abstraction methods on BioPAX aim to extract contextual information from the knowledge graph. In this work we study the abstraction of BioPAX to SIF (Simple Interaction Format), a binary interaction file format modeling interactions between proteins and chemicals. Paxtools is the only complete tool capable of handling BioPAX' complexity, and the only one (with its derivative tool ChiBE) able to abstract BioPAX to SIF using graph patterns based on SIF abstraction rules describing 14 biological configurations. However, SIF rules and Paxtools patterns are ambiguously documented, leading to misunderstandings. Here we explore SIF rules' formalization through a detailed comparison of their descriptions and Paxtools patterns codes. The result highlights a gap between SIF rules' descriptions and which patterns Paxtools extracts in the graph. Finally, we propose a transparent and FAIR BioPAX to SIF abstraction method introducing SPARQL queries recapitulating SIF rules. We tested these SPARQL queries by abstracting the PathBank and Reactome human pathway databases in BioPAX to SIF. SPARQL queries and Python code are available at: https://github.com/CecileBeust/BioPAX-To-SIF-SPARQL.git

Keywords

BioPAX, SIF, SPARQL

Poster summary

BioPAX (for Biological PAthway eXchange) is a standard Semantic Web format based on OWL (Ontology Web Language) designed for the representation of metabolic, signaling pathways, molecular and genetic interactions and gene regulation networks [1]. The BioPAX ontology provides classes and properties allowing to finely describe biological information using a controlled vocabulary. However, the counterpart of its high expressivity is an important complexity of the resulting BioPAX graphs, that hinders downstream analyses and reasoning tasks [2]. Overall, there is a lack of easy-to-use tools to work with BioPAX models, making its accessibility limited to a range of specialist users. There is thus an important need to exploit BioPAX' richness while accommodating its complexity. Abstraction methods on BioPAX models partially address this need by extracting contextual information from the knowledge graph while getting rid of part of its complexity.

Here, we focus on the abstraction rules for the conversion of BioPAX to SIF (Simple Interaction Format), a binary interaction file format allowing to represent interactions involving proteins and chemicals. The SIF format was introduced in the early 2000s for describing interactions in a biological network using the Cytoscape network visualization software [3]. SIF interactions are presented in tabular format with three columns: a first column for the source entity, a middle column for the interaction type, and a third column for the destination entity. This simplicity of representation makes the format easy-to-use, understandable by end-users and easily importable into a wide range of network visualization and analysis softwares. As for the tools available to convert BioPAX to SIF, there are few easy-to-use tools for working with BioPAX, and even fewer providing abstraction methods. The

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Paxtools Java library [4] is the only complete BioPAX API developed to date, capable of handling BioPAX' high intrinsic complexity and performing multiple tasks on BioPAX files (manipulation, validation and conversion). Paxtools, and its derivative visualization tool ChiBE [5], are the only tools able to abstract BioPAX models to SIF. The abstraction is done using inference rules that we will call SIF abstraction rules. These rules were defined by the SIF development team and are available on the PathwayCommons [6] website[1]. SIF abstraction rules describe 14 biological configurations, and summarize them in protein-protein, protein-chemical and chemical-chemical interactions covered by the SIF format. Technically, SIF abstraction rules are modeled in the Paxtools code with Java graph patterns. A pattern is a fixed number of BioPAX elements that satisfy a list of constraints [7]. However, the 14 SIF abstraction rules and their corresponding Paxtools patterns are ambiguously documented, potentially leading to errors or misunderstandings of the abstraction results.

In this work we investigated how SIF rules are described on PathwayCommons and in the Paxtools code. We highlighted that SIF abstraction rules are described at different levels, revealing three possible meanings of description for each one of the 14 rules: a diagram on the PathwayCommons website associated to a short textual description (meaning 1), a textual description in the Paxtools code (meaning 2) and the code of the Paxtools pattern modeling individual SIF rules (meaning 3). We explored the formalization of SIF abstraction rules through a detailed comparison of the three identified meanings of description of the rules, and highlighted the different constraints found in each meaning. The result highlights an important gap between the description of SIF rules presented to the user, and what SIF interactions are truly extracted by the Paxtools patterns in the BioPAX graph. We also investigated the potential errors generated by insufficient description of abstraction rules. We illustrated and quantified them with a biological example, by generating the three distinct SIF abstractions of the BioPAX Kidney function pathway from PathBank [8] obtained with the three meanings of description previously identified. Overall, SIF abstraction rules descriptions and documentation lack transparency, as evidenced by the three identified meanings of descriptions that give rise to three different types of SIF abstractions with distinct constraints not being explicited to the user, and that yield different results.

Our contribution is to propose a **transparent and FAIR BioPAX to SIF abstraction method** by introducing SPARQL queries recapitulating SIF abstraction rules at their three meanings of description. We developed a SPARQL query for each one of the 14 SIF abstraction rules, and for each one of the three meanings of description, highlighted the different SIF interactions that can be obtained with the three meanings. Each SPARQL query is transparent and documented with a clear textual description accompanied by a diagram of the BioPAX pattern extracted, and the SIF configuration obtained. As a case study, we generated using our SPARQL queries the SIF abstractions of the BioPAX exports of two large pathway databases: PathBank (version 2019, from PathwayCommons) and Reactome BioPAX Homo Sapiens (version 94, September 2025). The SPARQL queries scale well on such large BioPAX models. SPARQL queries and associated Python codes are available as Jupyter notebooks on GitHub at: https://github.com/CecileBeust/BioPAX-To-SIF-SPARQL.git

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Declaration on Generative AI

The author(s) have not employed any Generative AI tools.

[1] https://www.pathwaycommons.org/pc2/formats

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